# LOKLOK ??????
# ??????? ????? ???? ????

?? ????? ????????? ??????? ????? ??? ??????? ?? ????????
????????? ????? :???????? ???? ???????

????? ?????? ??? ???? ????
??? ?? ??????? ??????? ???????? ???? ??????? ???????
2026 ??????

???????? :??????? ????? ????? ???????? ?????? ??????? ??? ?????
.???????? ?????? ???????? ????????? ???????? ?????????? ?????????
????????? ??????? ???? .??????? ?????? ??? ???? ?????????? ??????
.????? ?????? ?????? ????? ??????? ???????? ??????

.42/42 :?????? .???? 359 :??????? ????????

# ???????? ?????? .1

??????? ????? ??? ????? ???? ?????? ??? ??? ???? ???????? ???? ?? ?????? ?????? ????? ??? ???????? ?????? ????? ?? ????????? ?Epistemic Systems Theory (EST) ????? ?????? .??????? ????? ??? ?????????? ?? ????? ??????? ???? :?????? ??? ??????? ?????? ?LOKLOK ???? ????? ?????? ??????? ?? ?????? ????? ?? ???? ???????? ??????? ?????? 52 ???????? ????????? ?????? ??? ????? ??????? ????? ?????? ??????? .????????? ????????? ???????? ?????? ???????

???? ??? ??????? ?? ???????? ??????? ????? ??? ?????? ??????? ?????? :????? ???? ???? ????? ????? ??? ????????? ???????? ??? .?????? ????? ??????? ?????? ??????? ????? ??? ???? 18? ?A/B/C ????? ?????? ?????? 500 ???????? ??????? .GRM-IRT ???? ?????? ????? ?? ??????? ?????? ?????? .?????? ????? ????? ???? ????? ??????? ?? ???? ????? ???? ??

(???? 18 ?????? 500) ???????? ????????? ???????

| ???????? | 95% ???? | ??????? | ?????? |
|---|---|---|---|
| $t(499) = 33.93$, $p < 0.001$, $d_z = 1.52$ | $[0.779, 0.875]$ | $+0.827$ | $\Delta$SR$_{\text{eval}}$ |
| $t(499) = 33.74$, $p < 0.001$, $d_z = 1.51$ | $[10.64, 11.95]$ | $+11.30$ | $\Delta$SR$_{\text{comp}}$ |
| $t(499) = -74.41$, $p < 0.001$, $d_z = -3.33$ | $[-0.188, -0.178]$ | $-0.183$ | $\Delta$ engagement |
| ??????? ?????? ??? ?????? ???? | HDI $[0.7793, 0.8764]$ | $+0.8265$ | Bayesian $\Delta$SR$_{\text{eval}}$ |

?? ??????? ??????? ????? .???? ???? ?? ??????? ?? ??????? ??????? ????? ?(0.6265+ ????? ???) ??????? ???? ???? (B) ???????? ?????? ???? ?(0.0121-) ??????? ????? ?? ?????? ERIC ????? ?(0.1905+) ???? ????? BRS ??? ?????? ??? ?? ???? 256 ??? ?? ???? ???? .0.3791+ ????? B$\times$C ???????? ?? 78.5 ?????? ??? ????????? ??????? ??? .(PRCC $+0.77$) cred_dbi ????????? ????? ??? ?(,0.383] [0.802 ??????? ????? ?0.567+ ??????) ?????? ????? ??? .???????? ??????? ????????? ??????? ???? ??????

???? ??? ??????? ?????? ??? ???? ?? :????? ?? ??? ???????? ????? ???? ??? ?v8 ??????? ?? ???? ???? ?????? ??? ?NMLS ?????? ???? ???? ?????? ??????? ???? ?????? ??? :?????? ???????? ?????? .?????? ?????

??????? ?????? 52??? Respondents .(0.827+ ????? 0.808+) ????????? ?? ???????? ?????? ?????? ?????????? ?????? ??????? ??? ????????? ?????? ?????? ????? (Gemini) ???? ????? ???????? ????????? ????? ????? .??????? ??? ?(0.163+) ??????? ????? ????? (2.650+) ???? ????? ????? ???????? .????? ?????? ????? ???? 18 ?????? ????? ?????? ???? ?????

## ????????? ???????? ??????? .2

??????? ??????? ????? ?????? ????? :???????? ?? ??? ?? ???????? ????? ??? ??????? ????? ????? ???? .?????? ???? ?? ???? ????? ???????? ?????? ????? ??????? ??????? ??????? ?????? ?????????? ??????? ????????? ??????? ??????? ??????? ?????? ??????? ??????? ???? .?????? ??? ???? ?? ?????? ????? ?????????? ?????? ???? ?????????? ??? .???????? ???? ???? ?????? ?????? ???? ????? ?????? ???????????

:?? ????? ?????

1. ?(EST-P1) ??????? ?????? ???????? ????????? ?????? ????? ????? ?? .

2. ?(EST-P2) ??????? ??????? ???????? ?????? ????? ????? ?? .

3. ?(EST-P3) ?????? ????????? ?????? ???????? ??????? ???? ?? .

4. ?(EST-P4) ?????? ???? ????? ?? EST ????? ?????? ?? .

5. ????????? ??????? ???? ???????? ???????????? ?? .

6. ????????? ??? ?????? ??????? ??????????? ??????? ???? ???? .

:???? ??? .?????? ??????? ????? ?????? ?????? ????? ?????? ???? ???????? BRS, MORE, ERIC, Echo Meter,) ??????? LOKLOK ????? ???????? ?????? EST ??????? ?????? ????? ???? ????? ????? ?(Algorithmic Cooling, Zewail gate ????? .????? ????? ?????? ?GRM-IRT ???? ????? ???????? ????? ?????? ???????? ???? ????????? ?? ?? ??? BGE-M3 ??????? ????? ???????? .?????? ???????? ?????? ?? ???? ??????? ?????? ?????????

# ????????? EST ??????? .3

??? ????????? ?????? :EST-P1 .??????? ??????? ????? EST ????? .??????? ??????? ??? ????? ???????? ?????? :EST-P2 .????????? ????? ???????? ?? ???????? ?????? :EST-P4 .???????? ??? ?????? :EST-P3 ???????? H3 ??????? ???? H2 ????????? ????????? H1 :????????? .???????? ??????? H6 ?(?????????) DeGroot ?????? H5 ???????? H4 ???????????? .(?????????) ???????? ?????? H7 ?(?????????) ??????

????????? ?????

| ????? | ???????? ?? ?????? | ??????? | ??????? |
|---|---|---|---|
| ????? ????? ??????? PB ??? ????? ???????? ???? ????????? ???????? ?? BRS ???? ?BRS? ????? ?????? ?????? ????? ??????? ??????? ?? ????????? ???????? | | H1, H4, H5 | EST-P1 |
| ???? ????? ?????? ?? MORE/Lagrange ???? ?????? ???????? ?Echo? ??????? ????? ??????? ???????? ??? ????????? ?? ????? ??????????? ??? ?????? ?????? | | H2 ?????? | EST-P2 |
| ?????? :????? ????? ?????? ????? ????? ???????? ?????? ??? ?????? ???? ????? ????? ????? ??? ???????? ????? | | H3 ?????? | EST-P3 |
| ???? ?????? ??? ??????? ????? ???? ???????? ????? ????? ?????? ?????? ????? ???????? ????? | | H7 ?????? | EST-P4 |

# ??????? ???????? .4

21 ?? ??????? ??? ???? .????? ????????? 60?? ????????? ?????? ???? ?????? ?????? ????? .??????? ????? 52 ???? ?????? ?????? ?????? ????? ???? ?? (PB) ??????? ??????? ??????? ?????? 11 ?? (BRS) ?????? ????? ??????? .?????? ????? ???????? ?????? ??????? ???? ??????? ?????? ????? ??????? ??? ?($\omega = .964/.962$ ?PB $\alpha = .947$ ?BRS $\alpha = .957$) ????? ??????? ????? ??? ?????????? ?????? ??? ???????? ????? .???? ??????? ??????? .??????? ?????? ??????? ????

185 ?? ???? ????? ??? ??????? 160 ?? ????? ?????? ?? ?????? ?????? ???? .????????? ?????? 25? ???? ????? 4,828? ??????? ?????? ???? 60? ??????? ??????? ???? 167 ???????? ?????? ????? ????? ??? 31 52??? ??????? ???? .??? 198

## ???????? Sentence-Transformer ???? 4.1

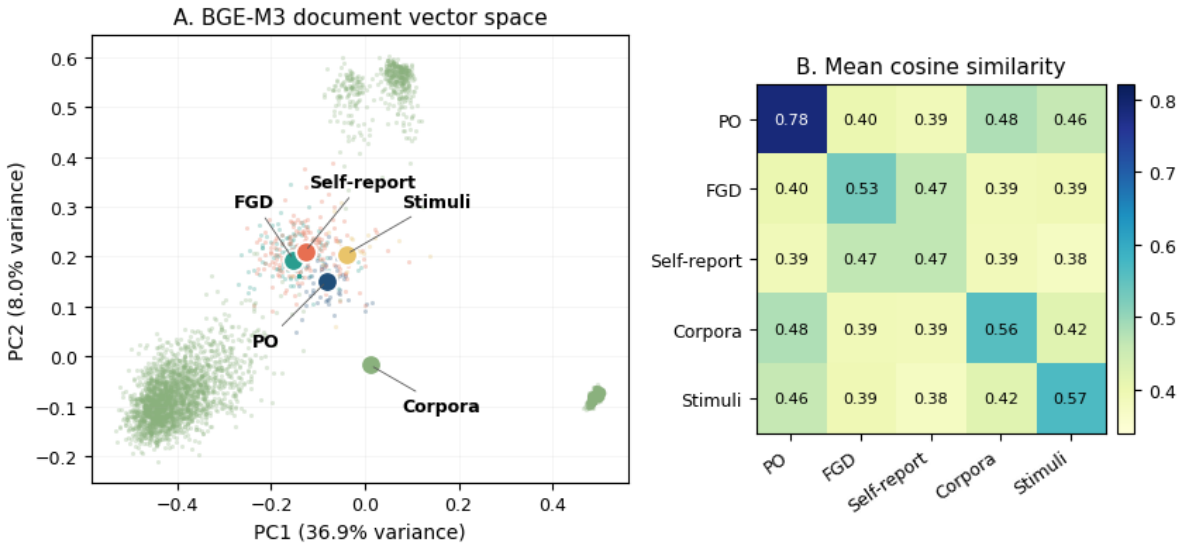
??? ????? ?????? ????? ??? ????? ??? ?BAAI/bge-m3 ?? ??????? ??????? ??? ?????? ????? 1,024 ?????? ?????? ?????? ???? .XLM-RoBERTa ???? ??????? ????? ?????? ??? ???? ?????? ????? ???????? ?????? .????? 8,192 ????? ?? (zero-shot): ????? ????? ??? ?????? ??????? .???????? ????? ?? .??????? ?? ?? ????? ????? ?? ???? ??? ????????? ?????? ???

???????? ?? ??? ??????? ?? ?? ????? ?????? 5,258 ???? ????? :??????? ??????? ????? ??????? ???? Colab T4 ??? ????? ????? ?? ??????? ?????? ???????? ??????? ??????? ??? ???? ??? ???????? ????? ?L2 ?????? ?????? 31 ?? ?????? ??? 167 ??? ?? ???????? ?????? 32 ??? ???????? ?????? ????? .???? API ???? ??? ??????? ?????? ??????? ???? ?????? ??????? .???? ??? .????? ????? 1,024??? ??????? ??? ?????? ???? ????? ?? ???

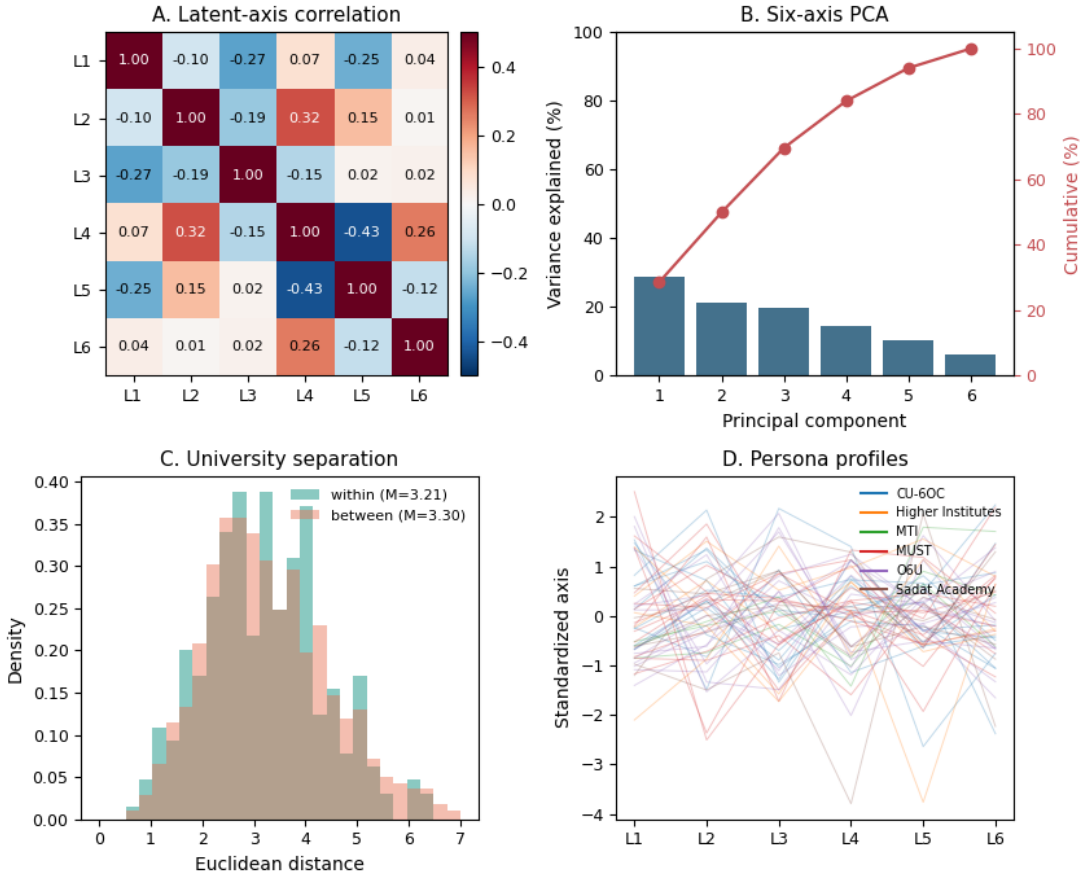
## ?????? ??????? ??????? ????? 4.2

??????? ?? 44.9% ??????? ???????? ???? .????? ????? 5,258 ???????? ?(0.785) ????????? ???????? ?? ?????? ???? ???? ????? ???? .(8.0%? 36.9%) ???????? ?(0.531) ??????? ??????? ?(0.559) ?????? ?(0.568) ???????? ?? ?????? ?????? ????? ?(0.48–0.38) ??? ??????? ??? ??????? .(0.466) ?????? .???? ?? ????? ??? :0.196 ???????

??????? ???? ??????? ?????? ???????? ???? ??????? ??????? ?????? ?????? ????? 4.815 ??????? ??? ??? ?????? 52??? .??????? ????? ?????? ????? .?????? ????? ?? 49.9% ??????? ???????? ???? .0.434 ????? ???? ??????? :(0.974 ??????) 3.295 ???????? ???? 3.211 ??????? ???? ??????? ?????? ???????? $Q_0$ ?? ?????? 500??? ????????? .??????? ???? ?? ??????? .?????? 500 ?? ?????? 52 ????




??????? .(?) ?????? ??????? ??????? (?) BGE-M3 ????? ?????? ???? :1 ????? .????? ????? ????? ???? ?????




????? ?(?) ??????? ???????? :????? 52??? ?????? ?????? ??????? :2 ????? .(?) ????????? ???????? ?(?) ???????? ???/???? ???????? ?(?) PCA

# ?????????? ??????? .5

PCA + LLE) ?????? ?????? ?? ?IPW? DML?? ?????? ????? ???????? ?? ????? ?DML ??? 0.042 ??? 0.887 ?? ???? ?????? ???? ????? .(+ Laplacian eigenmaps ??? ??? .?????? ?? ????? ??? ??????? ?(0.127) ???????? ??? IPW ????? ??? ????? ????? ???? ??) ????? KDE ????? .4.82 ??? ??????? ????? ??????? .????? 500 ?????? ?$Q_0$ ??????? (?????? ??? ??????? ???

???? ????? .???? 18 ??????? $2^3 = 8$ ???????? ??????? ??? ????? ?? ??????? Hamiltonian, Laplace, Lyapunov, DeGroot, Lagrange, :?????? ?????? ??? ?????? ??? ?????? ????? ???????? ???? ????? ????? .Hilbert, Percolation, Entropy ?? ????? 35?? Samejima GRM-IRT ?????? ???????? ?????? ?????? .???????? ?5.119 ?????? PB ????? ??? G0 ????? ??????? .MCS+PB ?? ???? 8? NMLS .52??? ?? ??????? ????? ?????? ????? ?????? ?????? ??????

????????? t ???????? ?????? ???????? ??????? .??????? ????????? ?????? ??????? ??????? ???????? ?95% ???????? ????? ?????? ?Cohen's $d_z$? ?????? ???? ????? ????? ????? ???? ??????? ???????? .Wasserstein ??????? ?????? ???????? ?????? ???? ???????? ???? ????? ????? ??? ????????? ?$P(\Delta > 0)$ ?????? ?? ????? 256 ???? .??????? ???????? ??? ??? ????? ??? .????? ?? ?????? ??? ???? ??

.????? 192 = ???? 8 $\times$ ????? ????? 24 ???? ?????? ????????? ????? ??????? ???? ?? ?????? .???? 18 ?????? ????? ?????? ???? ????? ???? ????? ??? .?????? ???? ???? ?? ??????? ?????

# ??????? .6

## ??????? ???????? ??????? ??????? 6.1

?7–1 ????? ??? 0.827+ ?????? ???? :????? ?????? ?? ???????? ??????? .1–0 ????? ??? 0.183- ????? ??????? ?100–0 ????? ??? 11.30+ ????? ????? .($d_z = -3.33$) ?????????? ????? ??????? ?????

A$\times$C ?A$\times$B $+0.0216$ ?C $-0.0121$ ?B $+0.6265$ ?A $+0.1905$ :??????? ??????? ????? ????? BRS ????? ?????? MORE .A$\times$B$\times$C $+0.0886$ ?B$\times$C $+0.3791$ ?$-0.0084$

??????? ??? ????? B×C ???? .???????? ?????? ??? ?????? ???? ERIC? .??????? ??? ????? ??????? ????? ???? ??? ??????

## ??????? ???? ?? 6.2

??????? ????? ??? Hilbert? Lagrange? DeGroot ?? ???????? ????? ????? ????? ,0.197] 0.233 ?????? $\triangle SR_{eval}$ ???? DeGroot ????? .???????? ??? ?? ??????? ?????? Hilbert ?????? ?[,0.713 0.787] 0.750 ?????? Lagrange ?????? ?[0.269 ????? ??? ?B ?????? ?????? ??????? ?? Lagrange .[,0.402 0.503] 0.453 ???? ??? ??????? ?????? ???? ??????? ???? ???? ????? ?? Lagrange-off .???? Lyapunov? ?????? ??? ??????? ???? Entropy? Percolation? Laplace .????? .??????? ???? ??? ?????? Hamiltonian?

????? .??????? ???? ???? ?????? ?????? ?? ??????? ????? ????? ???? ???? [,0.029- 0.010] 0.009- ?????? $\triangle SR$ ????? ???????? ?????? ?? ???? ????? ????? .[,0.039- 0.001] 0.019- ?????? ?????? ???????? ???????? ??? .(0.818+? 0.808+) ?????? ??????? ?? ??????? ??? ?????? ??????? ?????? ?????????? ????? ????????? ?????? ??????? 52??? ?????????? ???? .????? ????? ??????? ???????? ??????? .??????? ???????? ??????

??????-???? ????? ????? ?????????? ?????? ??? ????? ???? 256 ??? ?0.593+ ???????) 1.810+ ??? 0.244- ?? ??????? ?????? .????? ???? ??? ???? .????? ???????? ?? 96.5% ?(,0.383] [0.802 ??????? ????? ?0.567+ ?????? cred_dbi :??????? ????? ???????? ???? .78.5 ?????? ??? 0.827+ ???????? ???????? echo?? ??????? ?????? .0.27- cred_echo ?0.31+ cred_brs ?0.77+ ?????? ??? ???? ?????? ????? :????? ????? .???? ???? ????? :??????? ??? .??????? ????????? ???? ?????? ?? ??????? ???????? ?????? ???? ???????? ??? ?????? ????? ????????? ??????? ???? ??????? ??????? ?? .????? ???? ?????????

## ??????? ??????? ??????? 6.3

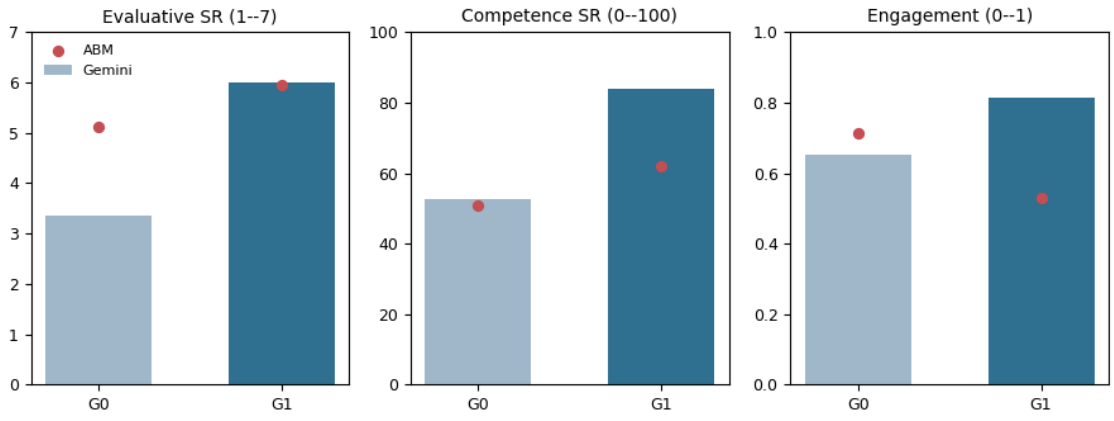
50??? ?(0.827+ ????? [,0.760 0.856] 0.808+) ???? 60 ???? ????? ??????? ?(???? 100% ?[,0.805 0.884] 95% ?0.839+ ???????) ??????? ?????? ????? ???? ??????????? ????? .(0.832+/0.826+/0.827+) ER/WS/BA ???????????

B ???? .??? ???? A? ?BRS ?????? ??? ??? ???? ??????? ????? :?????? ????? ???? ??? ?????? ???? ??????????? ???? ??????? ?? ????? ?????? .?????? ?????

????? ??????????? .??? 198? ???? 52 ?????? ?????? ???? ?????? ?????? ?583. ??? $R^2$ ???? ?????? ??? 167 ?????? ?($R^2 = .915$) ???? PB?? ????? ???????? ?????? ????? ?? ??????? ??? ?? .???? $R^2$ ?? ??? ??????? ???????? ????? ??????? ??? ?? ????? ??? ???? 167? ???? 52 ??? .?????? ???? ?????? ??????? ??? ?????? .???? ???? ?? ????? ??? ??????? ?? ??? ?????? ???? ??? ????? ??? ?1.69 ???? ??????? ???? ????? ?? 0.763 ???? ????? ??? ???? ?????? ?0.636+ ??? ΔSR ????? .?????? ????? ????? ?????? ???? ???????? .????? ??????? ????? ?[,0.261- 0.122-] 0.191- ???????

## ?????? ??????? ??????????? ??????? 6.4

[,2.342 2.967] 2.650+) ?????? ????? ???????? ?????? ????? ??????? ????? ????? ????? ????? ?(11.30+ ????? 31.15+) ??????? ???? ?????? ?(0.827+ ????? MAE :$SR_{eval}$?? ??????? ??????? ?????? ????? .(0.183- ????? 0.163+) ??????? ?????? ??????? .Spearman $\rho = 0.071$ ?Pearson $r = 0.269$ ?RMSE 0.842 ?0.655 ?? ????? ????? ?? ?????? ??????? ????? ?? ???? ?????? ?????? ??? ??? ?????? ????? ???????? .??????? ????? ???????? ???? ???? ????????? ????? .???? 18 ?????? ????? ????? ?????? ?????




??? ?????? ??????? ????? ?????? ??????? ??????????? ??????? :3 ????? ??? ??????? ?? ????????? ????? ??? ?????? ???????? ??????? .??????? .???????

## ?????? ?? ??? ?????? .7

?????? ?????? 52??? ?????? .?????? ?????? ????? ???????? ????? ?? NMLS ??? .?????? 500 ????? ??????? ????? 500??? ???????? .??????? ???? ???? ????? Hamiltonian .????? ??????? ???? ??? ???? ????? ????? ????????? ???? ???? ????? ????? BRS .????? ?? ??????? ?????????? ??????? ???? ??????? .v8 ???????? ????? ??? ????????? ????? ?????? .????? ???? ????? .??????? ??????? ??????? ??????? ????????

.????? ?????? ????? ????? ????????? ????? ??? ?????? ????????? ??????? ??? .????? ???????? ??????? ????? .??????? ?????? ??? ??????? ?????? ?????? ??????? ??? ?? ???????? ????? ??? ????? ??? ??????? ???? .???????? ??????? ????? ??? .??????? LOKLOK ??? ??? ?? ?????????

## ???????? ?????? ??????? .8

?? ??? ??????? ??????? ???? ????????? ?????? ?? ?? ??????? ????????? ??????? ??????? LOKLOK ????? ???????? ????? ??? EST ???? .???? ????? ???????? ????????? ??????? ????? ???????? ????? ??????? ?????? ????? ????????? ??????? ?????? ?? ???????? ???????? .?????? ??????? ????? ?? ???? EST ??????? ????? .??????? ??????? ???? ?????? ??? ?????? ????? .????????? ????? ???? ?????? ?????? ???????? ???????

?????? 0.183-? ?????? ???? 0.827+ ?? ????? ???? ??????? ??????? ????????? (?????) :???????? ??????? ???????? .??????? ????? ?? ????? ?????? 96.5%? ?? ???? ???? ?MORE ???? ??????? ??? ?????? ???? ???? Lagrange ????? ??? (??????) .????????? ??? ?? ?????? ??? ???? ?????? :??????? ????? ?????? ??????? ???? ????????? ???? ????? ?????? ??????? ???? ???? ?????? ??? ?????????? ??????? ??? ????? (??????) .??????? ?? ????????? ???? ????? (??????) .??? ???? ?????? ?????? ????? ??? ????? ?????? ?? 78 ???? ?????? ???? ????? ?????? ?? ??? 15% ??? ?????? ???????? ??????? .???????? ??????? ???? ?????? ??????? ????? ????